

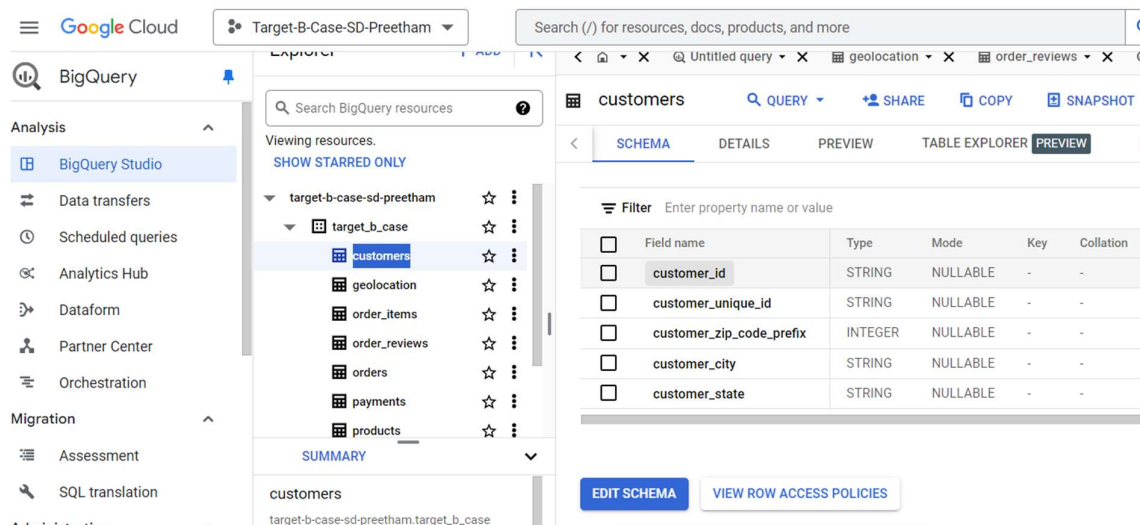
TARGET BUSINESS CASE

Problem Statement:

Assuming you are a data analyst/ scientist at Target, you have been assigned the task of analyzing the given dataset to extract valuable insights and provide actionable recommendations.

1. Import the dataset and do usual exploratory analysis steps like checking the structure & characteristics of the dataset:

- Data type of all columns in the "customers" table.



COLUMN

customer_id
customer_unique_id
customer_zip_code_prefix
customer_city
customer_state

DATA TYPE

STRING
STRING
INTEGER
STRING
STRING

2. Get the time range between which the orders were placed.

```
SELECT min(order_purchase_timestamp) as lower_limit,  
max(order_purchase_timestamp) as upper_limit  
FROM `target-b-case-sd-preetham.target_b_case.orders`
```

Query results

JOB INFORMATION		RESULTS	CHART	JSON	E
Row	lower_limit	upper_limit			
1	2016-09-04 21:15:19 UTC	2018-10-17 17:30:18 UTC			

3. Count the Cities & States of customers who ordered during the given period.

```
SELECT COUNT(DISTINCT customer_city) AS City,  
COUNT(DISTINCT customer_state) AS State  
FROM `target-b-case-sd-preetham.target_b_case.orders` AS o JOIN  
`target-b-case-sd-preetham.target_b_case.customers` AS c  
ON o.customer_id = c.customer_id;
```

Query results

JOB INFORMATION		RESULTS	CHART
Row	City	State	
1	4119	27	

2. In-depth Exploration:

1. Is there a growing trend in the no. of orders placed over the past years?

```
SELECT  
  EXTRACT(YEAR FROM order_purchase_timestamp) AS Year,  
  COUNT(DISTINCT order_id) AS total_orders  
FROM `target-b-case-sd-preetham.target_b_case.orders`  
GROUP BY Year  
ORDER BY Year;
```

Row	Year	total_orders
1	2016	329
2	2017	45101
3	2018	54011

Yes, there is a growing trend with the progression of years.

2. Can we see some kind of monthly seasonality in terms of the no. of orders being placed?

```
SELECT  
  EXTRACT(YEAR FROM order_purchase_timestamp) AS Year,  
  EXTRACT(MONTH FROM order_purchase_timestamp) AS Month,  
  COUNT(DISTINCT order_id) AS total_orders  
FROM `target-b-case-sd-preetham.target_b_case.orders`  
GROUP BY Year, Month  
ORDER BY Year, Month;
```

JOB INFORMATION		RESULTS	CHART	JSON
Row	Year	Month	total_orders	
12	2017	9	4285	
13	2017	10	4631	
14	2017	11	7544	
15	2017	12	5673	

Yes, there is a seasonality being observed in the number of orders placed. Certain months like November 2017, January 2018 have witnessed a high surge in the number of orders made.

Actionable Insights and Recommendations

Maximize on Seasonal Promotions

Insight: Order volumes peak in some months and fall in some suggesting seasonal data effects.

Recommendation: Use historical sales data and implement targeted marketing campaigns and promotions during peak shopping months to maximize sales.

1. During what time of the day, do the Brazilian customers mostly place their orders? (Dawn, Morning, Afternoon or Night)
 - 0-6 hrs : Dawn
 - 7-12 hrs : Mornings
 - 13-18 hrs : Afternoon
 - 19-23 hrs : Night

```
SELECT CASE
WHEN (EXTRACT(HOUR FROM order_purchase_timestamp)) BETWEEN 0 AND 6
THEN 'Dawn'
WHEN (EXTRACT(HOUR FROM order_purchase_timestamp)) BETWEEN 7 AND 12
THEN 'Morning'
WHEN (EXTRACT(HOUR FROM order_purchase_timestamp)) BETWEEN 13 AND 18
THEN 'Afternoon'
ELSE 'Evening' END as Day_Phase, COUNT(DISTINCT order_id) AS order_count
FROM `target-b-case-sd-preetham.target_b_case.orders`
GROUP BY Day_phase
ORDER BY order_count DESC ;
```

Query results

JOB INFORMATION		RESULTS	CHART	JSO
Row	Day_Phase	order_count		
1	Afternoon	38135		
2	Evening	28331		
3	Morning	27733		
4	Dawn	5242		

It is observed that most of the orders happen in the afternoon.

Evolution of E-commerce orders in the Brazil region:

1. Get the month on month no. of orders placed in each state.

SELECT

```

customers.customer_state,
EXTRACT(YEAR FROM orders.order_purchase_timestamp) AS year,
EXTRACT(MONTH FROM orders.order_purchase_timestamp) AS month,
COUNT(orders.order_id) AS total_orders
FROM `target-b-case-sd-preetham.target_b_case.orders` AS orders
JOIN `target-b-case-sd-preetham.target_b_case.customers` AS customers
ON orders.customer_id = customers.customer_id
GROUP BY customers.customer_state, year, month
ORDER BY year, month, customers.customer_state;

```

JOB INFORMATION		RESULTS	CHART	JSON	EXECUTION DETAILS	E
Row	customer_state	year	month	total_orders		
1	RR	2016	9	1		
2	RS	2016	9	1		
3	SP	2016	9	2		
4	AL	2016	10	2		
5	BA	2016	10	4		
6	CE	2016	10	8		
7	DF	2016	10	6		
8	ES	2016	10	4		
9	GO	2016	10	9		
10	MA	2016	10	4		

2. How are the customers distributed across all the states?

```
SELECT customer_state, COUNT(DISTINCT customer_id) as Population
FROM `target-b-case-sd-preetham.target_b_case.customers`
GROUP BY customer_state
ORDER BY Population desc;
```

Query results

JOB INFORMATION		RESULTS	CHART	JS
Row	customer_state	Population		
1	SP	41746		
2	RJ	12852		
3	MG	11635		
4	RS	5466		
5	PR	5045		
6	SC	3637		
7	BA	3380		
8	DF	2140		
9	ES	2033		
10	GO	2020		

Impact on Economy: Analyze the money movement by e-commerce by looking at order prices, freight and others.

1. Get the % increase in the cost of orders from year 2017 to 2018 (include months between Jan to Aug only).
You can use the "payment_value" column in the payments table to get the cost of orders.

```
WITH payments_2017 AS (
  SELECT
    ROUND(SUM(p.payment_value),2) AS total_payment_2017
  FROM
    `target-b-case-sd-preetham.target_b_case.payments` AS p
  JOIN
    `target-b-case-sd-preetham.target_b_case.orders` AS o
    ON p.order_id = o.order_id
  WHERE
    EXTRACT(YEAR FROM o.order_purchase_timestamp) = 2017
    AND EXTRACT(MONTH FROM o.order_purchase_timestamp) BETWEEN 1 AND
```

8

```

),
payments_2018 AS (
  SELECT
    ROUND(SUM(p.payment_value),2) AS total_payment_2018
  FROM
    `target-b-case-sd-preetham.target_b_case.payments` AS p
  JOIN
    `target-b-case-sd-preetham.target_b_case.orders` AS o
    ON p.order_id = o.order_id
  WHERE
    EXTRACT(YEAR FROM o.order_purchase_timestamp) = 2018
    AND EXTRACT(MONTH FROM o.order_purchase_timestamp) BETWEEN 1 AND
8
)

SELECT
  total_payment_2018,
  total_payment_2017,
  ROUND((((total_payment_2018 - total_payment_2017) / total_payment_2017),2) * 100
AS percentage_increase
FROM
  payments_2017, payments_2018;
2.

```

JOB INFORMATION		RESULTS	CHART	JSON
Row	total_payment_2018	total_payment_2017	percentage_increase	
1	8694733.84	3669022.12	137.0	

Monitor pricing strategies for increment in sales.

Insight: The percentage increase in cost of orders from 2017 to 2018 suggests sensitivity to price fluctuation.

Recommendation: Implement competitive pricing strategies considering the level of price sensitivity of customers.

2. Calculate the Total & Average value of order price for each state.

```

SELECT
  c.customer_state, ROUND(SUM(oi.price),2) as Total, ROUND(AVG(oi.price),2) as
Average
FROM `target-b-case-sd-preetham.target_b_case.order_items` oi JOIN
`target-b-case-sd-preetham.target_b_case.orders` o

```

```

ON o.order_id = oi.order_id
JOIN `target-b-case-sd-preetham.target_b_case.customers` c
ON o.customer_id = c.customer_id
GROUP BY c.customer_state
ORDER BY Total desc;

```

Row	customer_state	Total	Average
1	SP	5202955.05	109.65
2	RJ	1824092.67	125.12
3	MG	1585308.03	120.75
4	RS	750304.02	120.34
5	PR	683083.76	119.0
6	SC	520553.34	124.65
7	BA	511349.99	134.6
8	DF	302603.94	125.77
9	GO	294591.95	126.27
10	ES	275037.31	121.91

3. Calculate the Total & Average value of order freight for each state.

```

SELECT
c.customer_state, ROUND(SUM(oi.freight_value),2) as Total,
ROUND(AVG(oi.freight_value),2) as Average
FROM `target-b-case-sd-preetham.target_b_case.order_items` oi JOIN
`target-b-case-sd-preetham.target_b_case.orders` o
ON o.order_id = oi.order_id
JOIN `target-b-case-sd-preetham.target_b_case.customers` c
ON o.customer_id = c.customer_id
GROUP BY c.customer_state
ORDER BY Total desc;

```

Row	customer_state	Total	Average
1	SP	718723.07	15.15
2	RJ	305589.31	20.96
3	MG	270853.46	20.63
4	RS	135522.74	21.74
5	PR	117851.68	20.53
6	BA	100156.68	26.36
7	SC	89660.26	21.47
8	PE	59449.66	32.92
9	GO	53114.98	22.77
10	DF	50625.5	21.04

Actionable Insights and Recommendations

Insight: Certain states have consistently low order volumes or higher freight costs, indicating potential inefficiencies in inventory distribution.

Recommendation: Reassess inventory allocation and distribution strategies to ensure popular products are readily available in regions with high demand.

Analysis based on sales, freight and delivery time.

1. Find the no. of days taken to deliver each order from the order's purchase date as delivery time.

```
SELECT
  order_id,
  DATE_DIFF(order_delivered_customer_date, order_purchase_timestamp, DAY) AS
time_to_deliver,
  DATE_DIFF(order_delivered_customer_date, order_estimated_delivery_date, DAY) AS
diff_estimated_delivery
FROM
  `target-b-case-sd-preetham.target_b_case.orders`;
```

Row	order_id	time_to_deliver	diff_estimated_delivery
1	1950d777989f6a877539f5379...	30	12
2	2c45c33d2f9cb8ff8b1c86cc28...	30	-28
3	65d1e226dfaeb8cdc42f66542...	35	-16
4	635c894d068ac37e6e03dc54e...	30	-1
5	3b97562c3aee8bdedcb5c2e45...	32	0
6	68f47f50f04c4cb6774570cfde...	29	-1
7	276e9ec344d3bf029ff83a161c...	43	4
8	54e1a3c2b97fb0809da548a59...	40	4
9	fd04fa4105ee8045f6a0139ca5...	37	1
10	302bb8109d097a9fc6e9cefc5...	33	5

2. Find out the top 5 states with the highest & lowest average freight value.

```
WITH avg_freight_statewise AS (
  SELECT c.customer_state, ROUND(AVG(oi.freight_value),2) AS Average_Freight
  FROM `target-b-case-sd-preetham.target_b_case.customers` c JOIN
  `target-b-case-sd-preetham.target_b_case.orders` o
  ON c.customer_id = o.customer_id
  JOIN `target-b-case-sd-preetham.target_b_case.order_items` oi
  ON o.order_id = oi.order_id
  GROUP BY c.customer_state)
```

)

Lowest 5 states

```
SELECT customer_state, Average_Freight
FROM `avg_freight_statewise`
ORDER BY Average_Freight
LIMIT 5;
```

Row	customer_state	Average_Freight
1	SP	15.15
2	PR	20.53
3	MG	20.63
4	RJ	20.96
5	DF	21.04

Highest 5 states

```
SELECT customer_state, Average_Freight
FROM `avg_freight_statewise`
ORDER BY Average_Freight desc
LIMIT 5;
```

Row	customer_state	Average_Freight
1	RR	42.98
2	PB	42.72
3	RO	41.07
4	AC	40.07
5	PI	39.15

3. Find out the top 5 states with the highest & lowest average delivery time.

```
WITH avg_delivery_time AS (
SELECT
c.customer_state, ROUND(AVG(DATE_DIFF(o.order_delivered_customer_date,o.order_purchase_timestamp ,DAY)),2) AS delivery_time
FROM `target-b-case-sd-preetham.target_b_case.customers` c JOIN
`target-b-case-sd-preetham.target_b_case.orders` o
ON c.customer_id = o.customer_id
```

```
GROUP BY c.customer_state
```

)

Lowest 5

```
SELECT customer_state,delivery_time  
FROM `avg_delivery_time`  
ORDER BY delivery_time  
LIMIT 5;
```

Row	customer_state	delivery_time
1	SP	8.3
2	PR	11.53
3	MG	11.54
4	DF	12.51
5	SC	14.48

Highest 5

```
SELECT customer_state,delivery_time  
FROM `avg_delivery_time`  
ORDER BY delivery_time desc  
LIMIT 5;
```

Row	customer_state	delivery_time
1	RR	28.98
2	AP	26.73
3	AM	25.99
4	AL	24.04
5	PA	23.32

Actionable Insights and Recommendations

Improve Delivery Speed

- **Insight:** Analysis reveals delivery times differ across states, with some states experiencing longer delays.
- **Recommendation:** Enhance logistics partnerships to streamline the delivery process.

Find out the top 5 states where the order delivery is really fast as compared to the estimated date of delivery.

```
WITH avg_fast_delivery AS (
  SELECT
    c.customer_state, ROUND(AVG(DATE_DIFF(o.order_estimated_delivery_date, o.order_delivered_customer_date, DAY)), 2) AS Average_Fast_Delivery
  FROM `target-b-case-sd-preetham.target_b_case.customers` c JOIN
    `target-b-case-sd-preetham.target_b_case.orders` o
  ON c.customer_id = o.customer_id
  WHERE o.order_estimated_delivery_date > o.order_delivered_customer_date
  GROUP BY c.customer_state
)
SELECT customer_state, Average_Fast_Delivery
FROM `avg_fast_delivery`
ORDER BY Average_Fast_Delivery
LIMIT 5;
```

Row	customer_state	Average_Fast_Delivery
1	SP	11.17
2	ES	12.34
3	MS	12.41
4	DF	12.42
5	SC	12.51

Analysis based on the payments:

1. Find the month on month no. of orders placed using different payment types.

```
SELECT EXTRACT(YEAR from order_purchase_timestamp) as YEAR,
  EXTRACT(MONTH from order_purchase_timestamp) as MONTH,
  payments.payment_type,
  COUNT(DISTINCT payments.order_id) as total
FROM `target-b-case-sd-preetham.target_b_case.orders` AS orders JOIN
  `target-b-case-sd-preetham.target_b_case.payments` AS payments
ON orders.order_id = payments.order_id
GROUP BY YEAR, MONTH, payments.payment_type
ORDER BY YEAR, MONTH, payments.payment_type;
```

JOB INFORMATION	RESULTS	CHART	JSON	EXECUTION DETAILS	EX
Row	YEAR	MONTH	payment_type	total	
1	2016	9	credit_card	3	
2	2016	10	UPI	63	
3	2016	10	credit_card	253	
4	2016	10	debit_card	2	
5	2016	10	voucher	11	
6	2016	12	credit_card	1	
7	2017	1	UPI	197	
8	2017	1	credit_card	582	
9	2017	1	debit_card	9	
10	2017	1	voucher	33	

2. Find the no. of orders placed on the basis of the payment installments that have been paid.

SELECT

payments.payment_installments as Payment_Installment,

COUNT(DISTINCT payments.order_id) as total

FROM `target-b-case-sd-preetham.target\_b\_case.orders` AS orders JOIN

`target-b-case-sd-preetham.target\_b\_case.payments` AS payments

ON payments.order_id = orders.order_id

GROUP BY payments.payment_installments

ORDER BY total desc;

Row	Payment_Installment	total
1	1	49060
2	2	12389
3	3	10443
4	4	7088
5	10	5315
6	5	5234
7	8	4253
8	6	3916
9	7	1623
10	9	644

Tailor Payment Options

Insight: Payment modes trends reflect that there are specific preferences of a customer for specific types of payments and installment options for different customer segments.

Recommendation: Flexible payment options-certainly would likely appeal to customers with a wide range of different needs, thereby enhancing customer satisfaction and converting them into higher-income segment customers.

Conclusion

These actionable insights and recommendations implemented may greatly enhance the operational efficiency of Target in Brazil and improve its performance in terms of customer satisfaction. Through the use of data-driven strategies, Target will be able to meet better the needs of the target customers, thereby enhancing sales and even more so ensuring brand loyalty in a competitive environment.